Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-/ML-Assisted Advanced Plant Cell–Based Bio-Manufacturing Platforms			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		*Bio・Medical*	
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes / Bio-Equipment	*Bio-process Technologies*	*Industrial Biotechnology*	*Bio-cosmetic Materials*

Concept of Proposed Technology

Industrial Background and Technological Gap

With regard to the **UN Sustainable Development Goals (UN SDGs)**, the global **bio-industry** (bioeconomy; including food, cosmetics and pharmaceuticals) is undergoing a structural transition from **agriculture-dependent extraction models** toward **precision bio-manufacturing platforms** capable of producing high-value biomolecules with improved consistency, sustainability, and scalability. In this emerging landscape, technological competitiveness is increasingly defined not by access to biological resources, but by the ability to **design, control, and scale bio-manufacturing systems**.

Traditional plant-based extraction approaches suffer from inherent limitations, including **seasonal variability, low target compound abundance, land-use pressure, and environmental burden**. These constraints are particularly critical for **rare bioactive compounds** required in nutraceutical, cosmeceutical, and pharmaceutical applications.

Although plant cell–based fermentation offers clear advantages in terms of sustainability and supply stability, its **industrial penetration remains limited**. A major obstacle is the frequently observed **100–1,000-fold productivity drop** when processes are transferred from laboratory-

scale experiments to industrial-scale production. This gap cannot be closed through empirical optimization alone.

Artificial Intelligence (AI) / Machine Learning (ML) approaches have demonstrated strong potential for **adaptive bioprocess optimization**, especially in complex, multivariable systems. However, sustainable and transferable productivity gains can only be achieved when **AI-/ML-based process optimization is combined with biologically informed modulation of plant cell metabolism**, such as through elicitation strategies, feeding regimes, and controlled cultivation conditions.

While **Fraunhofer IME** has in-depth know-how in **fermentation process engineering and scale-up, Korean SMEs** have strong applied experience in working with **plants, plant cell systems, and plant cell metabolism in industrial contexts**.

Integrated Technology Positioning

The proposed technology integrates **applied plant cell bio-manufacturing expertise** with **advanced systems modeling, scale-up design, and industrial biotechnology validation**. This integration enables a transition from **laboratory-level reproducibility to industrial-scale reliability**, positioning the platform as a practical and future-ready solution for advanced bio-manufacturing.

Rather than offering a single optimized process or product, the proposed technology establishes a **transferable platform** that can be adapted to multiple plant cell systems and target compounds, forming a robust foundation for sustained innovation in the Korean bio-industry.

Core Concept

The core concept of the proposed technology is a **plant cell–based bio-manufacturing platform** that integrates **metabolic pathway modulation** with **AI-/ML-assisted bioprocess modeling**, enabling the **reproducible, scalable, and sustainable production of high-value bio-functional materials**.

Unlike conventional approaches that treat **metabolic optimization** and **process control** as separate challenges, the proposed platform unifies **biological pathway modulation** and **process-level optimization** within a single, coherent framework. This integration allows **predictive and adaptive control across scales**, from laboratory screening to industrial **stirred-tank bioreactor (STR) operation**, while preserving biological relevance and industrial robustness.

Main Functions

The proposed technology performs the following key functions:

- **Precision modulation of primary and secondary metabolite pathways**, allowing enhancement of target compound production while maintaining biological stability and industrial applicability.
- **Predictive control of plant cell growth and productivity** through hybrid (white-box/black-box) AI/ML models that link nutrient utilization, metabolic activity, and process conditions.
- **Stable and reproducible operation in industrial STRs**, enabling consistent performance during scale-up and reducing batch-to-batch variability.

Collectively, these functions address the **central bottleneck of plant cell fermentation**, namely the loss of productivity and reproducibility during scale-up, by enabling **data-driven yet biologically grounded process control**.

Potential Uses

The platform is designed for broad applicability across multiple **high-value bio-industry sectors**:

- **Nutraceuticals:** functionally validated food and nutrition materials derived from plant cell cultures with consistent quality and traceable production.

	• **Cosmeceuticals:** cosmetic ingredients developed with pharmaceutical-level scientific rigor, supporting efficacy-driven and sustainability-focused product concepts. • **Pharmaceuticals (future):** plant-derived, drug-relevant materials that may serve as starting points for future pharmaceutical development, subject to subsequent regulatory pathways. This multi-sector applicability ensures that the platform remains **technology-driven rather than product-limited**, supporting long-term industrial relevance and enabling participation by a wide range of Korean SMEs.
Rationale for Joint R&D Project	For the **global bio-industry** (bioeconomy), the industrialization of **plant cell–based bio-manufacturing** represents a strategically important yet technically demanding opportunity. While plant cell platforms offer clear advantages in terms of sustainability, supply stability, and access to rare bio-functional materials, their **translation from laboratory-scale feasibility to industrial-scale reliability** remains limited. Key barriers include **significant productivity losses during scale-up, batch-to-batch variability, and the absence of sufficiently predictive and transferable process control frameworks**. Overcoming these challenges requires a **convergence of applied biological expertise and advanced process engineering capabilities**. In particular, the successful industrial implementation of plant cell–based production systems demands the integration of biological pathway understanding, data-driven process modeling, and staged validation at pilot and pre-industrial scales. No single organization can efficiently cover the full spectrum of competencies, from biological system development to industrial-scale process validation. The proposed joint R&D project is therefore designed as a **collaborative framework** that brings together the capabilities of a **leading European Research and Technology Organization (RTO)** and **industrially oriented Korean small and medium-sized enterprises (SMEs)**. In this framework, the RTO contributes advanced know-how in **systems modeling, AI-/ML-assisted process optimization, scale-up design, and industrial biotechnology validation**, while Korean SMEs are expected to contribute **application-driven biological systems, applied bioprocessing expertise, and industrial translation capability** in subsequent phases of the project. Korea is particularly well suited for such a collaboration. The Korean bio-industry has demonstrated a **strong track record in the industrial exploitation of plant-derived bio-materials**, supported by rapid scale-up capability, advanced manufacturing infrastructure, and **global market leadership in cosmeceutical and nutraceutical sectors**. Compared to countries that emphasize primarily academic or craft-oriented approaches, the Korean bio-industry effectively combines **metabolic pathway analysis, applied bioprocessing, and fast market translation**, making it an ideal partner for **platform-oriented industrial R&D** in cooperation with international research institutes. Public support is essential to **de-risk this platform-level development**, particularly during the critical transition from **Technology Readiness Level (TRL) 3–4 to TRL 6–7**, where technical uncertainty and financial exposure are highest and private investment incentives remain limited. By enabling true co-development at pilot and pre-industrial scale, public funding ensures that the project outcomes extend beyond individual partner interests and contribute to the **long-term competitiveness, sustainability, and technological depth of the Korean bio-industry**.
Key Words	1. Plant Cell–Based Bio-Manufacturing Platforms 2. AI/ML-Assisted Bioprocess Modeling and Optimization 3. Primary and Secondary Metabolite Modulation 4. STR Scale-Up and Precision Fermentation 5. Cosmeceutical, Nutraceutical, and Pharmaceutical Bio-Materials

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (6-7)

R&D Objectives

The project directly addresses the critical gap between laboratory-scale feasibility and industrial-scale reliability that has limited the widespread adoption of plant cell–based bio-manufacturing in the bio-industry (bioeconomy). Specifically, the objective of this project is to establish an **industrial STR platform for plant suspension cell cultures**, integrating **AI-/ML-assisted hybrid process modeling** to enable **reproducible, predictable, and scalable production** from **1 L bench scale to 50 L and 500 L pilot scale**.

Importantly, the R&D objectives are **intentionally defined at the platform level**, rather than being limited to a single plant species, metabolite, or end product. This ensures that the technologies developed within the project are **transferable across multiple plant cell lines, target compounds, and application areas**, including nutraceutical, cosmeceutical, and future pharmaceutical domains. By focusing on platform robustness and reproducibility, the project aims to generate outcomes that remain valuable beyond individual case studies.

To achieve this overarching goal, the following **specific R&D objectives** are defined:

- **Development of hybrid AI-/ML modeling frameworks** that integrate mechanistic (white-box) models with data-driven (black-box) approaches to describe and predict **biomass formation, nutrient utilization, and metabolite production** in plant suspension cell cultures under industrially relevant conditions.
- **Extension of the hybrid modeling framework to secondary metabolite modulation**, incorporating elicitation strategies and cultivation inputs that influence pathway fluxes, thereby enabling rational optimization of target compound productivity.
- **Validation of reproducible performance across scales**, including intermediate-scale STR operation (**1–5 L**) and pilot-scale demonstration (**50–500 L**), to confirm model predictability, process stability, and batch-to-batch consistency during scale-up.
- **Establishment of transferable and regulation-aware process modules**, including defined operating windows and critical process parameters, that support **future industrial adoption, technology transfer, and downstream development**, while remaining adaptable to different plant cell systems and application requirements.

Collectively, these objectives ensure that the project delivers not only incremental performance improvements, but also a **coherent, scalable, and industry-ready bio-manufacturing platform** capable of supporting long-term innovation in the Korean bio-industry.

R&D Target Specifications

By addressing technical, economic, environmental, and standardization aspects in an integrated manner, the following target specifications define a **balanced and industry-relevant performance envelope** for the proposed joint R&D project. The targets are designed to reflect **industrial feasibility and scalability**, rather than optimized laboratory performance.

Technical Target Specifications

From a technical perspective, the project aims to achieve **quantifiable and industrially meaningful performance improvements** that directly address the key bottlenecks of plant cell–based bio-manufacturing. The following technical targets are defined:

- Achievement of a **≥2-fold improvement in volumetric productivity**, benchmarked against validated baseline processes, demonstrating that productivity gains are maintained during scale-up from laboratory to pilot scale, rather than being limited to small-scale experimentation.

- Reduction of **batch-to-batch variability by ≥30%**, reflecting improved process robustness, reproducibility, and controllability under industrially relevant operating conditions. This target directly addresses one of the major barriers to wider industrial adoption of plant cell fermentation technologies.
- Development and validation of a **hybrid AI-/ML-assisted process model** capable of reliably predicting biomass formation and secondary metabolite production at **50 L and 500 L STR scales**, thereby demonstrating predictive performance and scalability beyond small-scale experimentation.

These technical targets are intentionally selected to **ensure that the project outcomes are directly transferable to real-world manufacturing environments and relevant to industrial decision-making**.

Economic and Environmental Target Specifications

From an economic and environmental standpoint, the project targets measurable improvements that support sustainable industrial deployment:

- Demonstration of a **30–50% reduction in Cost of Goods (CoG)** potential, achieved through increased productivity, improved process stability, and reduced batch failures, thereby enhancing the economic attractiveness of plant cell–based bio-manufacturing
- **Life-Cycle Assessment (LCA)–verified reductions** in **land use, water consumption, and greenhouse gas (GHG) emissions** compared to conventional field-based sourcing, supporting the role of plant cell platforms as environmentally sustainable alternatives for producing high-value bio-functional materials.

These targets ensure that technological advances are aligned with **economic viability and environmental responsibility**, consistent with the principles of sustainable industrial innovation promoted by the TIPA program.

Standardization and Quality Perspective

From a standardization and quality perspective, the project aims to establish **process and documentation practices compatible with international quality expectations**, while remaining within the scope of a pre-competitive R&D project:

- Development of **Standard Operating Procedures (SOPs)** aligned with relevant **ISO and HACCP frameworks**, with a focus on process consistency, traceability, and quality awareness.
- Emphasis on **regulatory awareness rather than formal certification**, ensuring that the developed processes are structured in a way that facilitates future industrial qualification, technology transfer, and downstream regulatory engagement without prematurely committing to certification requirements.

R&D Plan

Overall R&D Approach

The R&D plan follows a **stepwise, data-driven development pathway**, ensuring continuity between biological experimentation, model development, and scale-up validation.

- Develop AI/ML models describing plant cell primary and secondary metabolism,
- Integrate these models into an existing AI/ML-based cultivation-condition model, and
- Validate the extended hybrid model through staged scale-up and pilot-scale operation.

WP1 (Months 1–12): Plant Cell Line Establishment and High-Throughput Data Generation

Objective: Establish a robust biological and data foundation for subsequent modeling activities. Establishing, characterizing, and banking of high-performing plant suspension cell lines.

- High-throughput cultivation in microtiter plate (MTP) systems
- Generation of datasets describing cell growth, nutrient consumption, and metabolite formation under systematically varied cultivation and elicitation conditions.

Outcome: Curated, high-quality datasets suitable for AI/ML model training and validation, with biological variability well characterized.

WP2 (Months 6–18): Hybrid AI/ML Model Development for Plant Cell Primary and Secondary Metabolism

Objective: Development of AI/ML models describing plant cell metabolism. Development of models representing primary metabolism (growth, biomass formation)

- Extension to secondary metabolite formation, using metabolomic and cultivation data
- Validation of model robustness using independent datasets

Outcome: Validated AI/ML models capturing the relationship between metabolic behavior and cultivation inputs.

WP3 (Months 10–20): Integration into Hybrid AI/ML Cultivation-Condition Model

Objective: Integration of the models developed in WP2 into an existing AI/ML-based cultivation-condition model.

- Integration of plant cell metabolic models with cultivation-condition models that describe nutrient distribution, oxygen transfer, and hydrodynamics
- Development of a hybrid AI/ML model linking metabolism, growth, and process conditions
- Model validation under controlled laboratory and bench-scale conditions

Outcome: A predictive, hybrid AI/ML model capable of simultaneously optimizing cell growth and secondary metabolite production.

WP4 (Months 12–24): STR Scale-Up and Pilot Validation (1–5 L → 50–500 L)

Objective: Translate the model-guided insights into pilot-scale validation.

- Transfer of optimized cultivation conditions into 1–5 L STR systems
- Scale-up and validation in 50–500 L STR
- Evaluation of process stability, reproducibility, and batch-to-batch variability

Outcome: Experimental validation that the hybrid AI/ML model remains predictive and robust under industrially relevant STR conditions.

WP5 (Months 24–30): Techno-Economic Assessment and Technology Transfer Preparation

Objective: Analyze industrial relevance and future exploitation readiness.

- Cost-of-Goods (CoG) analysis based on pilot-scale data
- Life-Cycle Assessment (LCA) to evaluate environmental performance
- Preparation of technology transfer packages, including model descriptions and process documentation

Outcome: A technology-ready and transfer-oriented platform, suitable for downstream adoption by Korean SMEs in subsequent project phases.

Overall R&D Plan Perspective

Together, these **WPs form a coherent, risk-mitigated R&D trajectory** that aligns with the TIPA 30-month execution constraint. By centering the project on AI/ML-based model development and

	integration, followed by focused pilot-scale validation, the **R&D plan ensures scientific depth, technical feasibility, and industrial relevance**, without overextending the project scope.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Fraunhofer IME** will serve as the **co-leading technical partner** in the joint R&D project providing core research, engineering, and validation capabilities that are essential for translating laboratory-scale plant cell processes into **scalable and industrially relevant bio-manufacturing platforms.**
	
	The expected roles and responsibilities of Fraunhofer IME include the following:
	
	• **Hybrid AI/ML Modeling and Systems Integration**
	  ○ Lead the development, integration, and validation of **hybrid AI/ML modeling frameworks** that combine mechanistic (white-box) models with data-driven (black-box) approaches.
	  ○ Ensure that the models reliably link cultivation conditions, **nutrient utilization, cell growth, biomass formation, and secondary metabolite production** under industrially relevant operating conditions.
	  ○ Integrate newly developed **plant cell primary and secondary metabolism models** into existing AI/ML-based cultivation-condition models, enabling joint optimization of biological and process parameters.
	
	• **Bioprocess Engineering and STR Scale-Up**
	  ○ Design, implement, and validate **STR scale-up strategies**, covering laboratory and intermediate scales (**1–5 L**) and pilot-scale validation (**50-500 L STR**), in line with the TIPA project duration
	  ○ Apply model-guided process optimization to support **stable and reproducible operation during scale-up, with a focus on process robustness and predictability rather than extensive process intensification**
	
	• **Provision of Research Infrastructure and Resources**
	  ○ Provide access to **pilot-scale bioprocessing infrastructure**, including STR systems, advanced monitoring and control equipment, and supporting analytical and computational resources required for model validation and scale-up studies.
	  ○ Contribute specialized expertise in **AI-/ML-assisted bioprocess optimization, systems modeling, and scale-up methodology**, drawing on Fraunhofer IME's RTO capabilities.
	
	• **Validation, Quality, and Regulatory Awareness Support**
	  ○ Conduct **pilot-scale validation studies** to confirm model predictability, process stability, and batch-to-batch consistency under industrially relevant conditions.
	  ○ Support the development of **quality- and regulation-aware process documentation**, including standardized operating concepts and data structures that facilitate future industrial adoption and downstream regulatory engagement, without committing to formal certification at this stage.
	
	Through these activities, Fraunhofer IME will ensure that advanced modeling and process engineering approaches are effectively translated into **reproducible, scalable, and industry-ready bio-manufacturing solutions**, forming a critical technical backbone of the joint R&D project under the TIPA framework.

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Expected R&R of Korean SME Partner	The Korean SME partner will act as the **industrially oriented development partner**, providing applied biological know-how, development resources, and industry-driven inputs that are essential for translating joint R&D outcomes into **scalable and application-relevant bio-manufacturing platforms**.
	
	The expected roles and responsibilities of the Korean SME partner include the following:
	
	• **Plant Cell Line Development and Biological System Management**
	  ○ Establish, characterize, and **bank high-performing plant suspension cell lines**, including documentation of baseline growth behavior, productivity, and stability.
	  ○ Ensure the availability of well-defined biological starting materials for modeling, scale-up, and pilot validation activities conducted within the project.
	
	• **Elicitation and Cultivation Input Design**
	  ○ Design and evaluate **elicitation strategies** (chemical and/or physical) and cultivation input concepts aimed at enhancing secondary metabolite productivity.
	  ○ Define practical **elicitation windows, feeding regimes, and operational parameters** from an industrial applicability perspective, providing essential input for model development and process optimization.
	
	• **Analytical Support, Testing, and Validation**
	  ○ Conduct **product analytics and quality characterization**, including HPLC and LC-MS analyses, to quantify target compounds and assess batch-to-batch variability across scales.
	  ○ Provide **validated analytical datasets** to support hybrid AI/ML model calibration, verification, and performance benchmarking in close collaboration with the research institute.
	
	• **Industrial Translation and Commercialization-Oriented Contributions**
	  ○ Translate R&D outcomes into **application-driven requirements** for nutraceutical, cosmeceutical, and potential future pharmaceutical domains, including target specifications and quality expectations.
	  ○ Contribute to **IP awareness, market relevance assessment, and future exploitation planning**, ensuring that project outputs are aligned with industrial feasibility and potential downstream adoption by Korean SMEs.
	
	Through these contributions the Korean SME partner will ensure that the jointly developed technologies are **industrially grounded, application-relevant, and transferable**, complementing the research institute's modeling and process-engineering leadership and supporting the development of a **robust, industry-ready plant cell–based bio-manufacturing platform**.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Overall Impact** Overall, this joint R&D project will deliver a **scalable, reproducible, and regulation-aware plant cell–based bio-manufacturing platform** that effectively bridges the gap between **laboratory innovation and industrial deployment**. By combining **industrial implementation capabilities of Korean SMEs** with Fraunhofer IME's research and engineering expertise, the project will generate **lasting technological capabilities, industrial know-how, and collaborative networks that extend beyond the project duration**. As a result, the project is expected to strengthen the position of the **Korean bio-industry as a globally competitive and sustainable key sector**, capable of addressing high-value international

markets while supporting long-term economic growth, environmental responsibility, and international cooperation, in line with the objectives of the TIPA program.

Technological Impact

From a technological perspective, the joint R&D project is expected to deliver **robust and transferable technological assets** that directly address the key limitations of plant cell–based bio-manufacturing identified in the earlier sections of this proposal.

- Establishment of a **validated and scalable plant cell bio-manufacturing platform**, capable of maintaining productivity, stability, and reproducibility during scale-up from laboratory to pilot-scale operation.
- Development of **transferable AI-/ML-enabled process modules** that integrate biological insight with predictive process control, allowing application to multiple plant cell lines, target compounds, and production scenarios.

Collectively, these outcomes will provide a **technology foundation that goes beyond single-use processes**, supporting long-term innovation and adaptability within the bio-manufacturing ecosystem.

Industrial and Economic Impact

At the industrial and economic level, the project will contribute to strengthening **the competitiveness and resilience of the Korean bio-industry, particularly in high-value segments.**

- Reduction of **structural dependence on imported bio-materials** through the introduction of sustainable, plant cell–based production routes.
- Creation of **high-skilled employment opportunities** in areas such as bioprocess engineering, data-driven process optimization, and quality management, contributing to SME capacity building.
- Enhancement of **SME competitiveness by enabling access to internationally validated, platform-based production technologies** that would otherwise be difficult to develop independently within typical SME R&D budgets.

Through these effects, the project supports the development of **knowledge- and technology-driven value creation, rather than volume-oriented** commodity production.

Sustainability and Societal Impact

In alignment with the **UN SDGs**, the project promotes sustainable and responsible industrial development through technological innovation.

- Implementation of **environmentally sustainable, production routes** that reduce land use, water consumption, and overall environmental burden compared to conventional agricultural sourcing.
- Support for **societal acceptance and regulatory compatibility** through reproducible, transparent, and well-documented bio-manufacturing processes
- Strengthening of **Korea–Germany collaboration in advanced bio-manufacturing** through structured knowledge exchange, joint technical development, and long-term partnership frameworks.

These outcomes contribute to the broader transition toward **sustainable bio-based industries**, delivering positive environmental and societal effects without introducing new regulatory or ecological claims beyond those already established in the proposal.

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